Pavit Thakur

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Education

Texas A&M University – *BS in Electrical Engineering (3.81) – Dean's Honor Roll* **May 2028**

Experience

Power Distribution Lead, Texas A&M Solar Car Racing **Aug 2025 – Present**

- Created a synchronous 12-5V buck converter in Altium Designer rated for 4A with 96% peak efficiency, bridging a high-voltage power supply with low-voltage communication systems.
- Established a CAN-BUS communication protocol on a MITSUBA motor controller to retrieve the RPM, motor current, and other telemetry data from the motor.
- Designed and soldered the 2-layer Driver Control Systems (DCS) PCB which connected CAN communication lines, throttle, the SPI module, cruise control, lighting, and various other control IO.
- Served as Co-Chief Electrical for the 2026 competition, leading the car through electrical and array scrutineering at FSGP, including battery protection checks for over/under temperature, current, and voltage

Avionics Engineer, Texas A&M Sounding Rocketry Team **Aug 2026 – Present**

- Engineered a 4-layer live telemetry PCB around an STM32 MCU, implementing a power delivery network with a 1A eFuse, USB-C connectivity, and low-inductance decoupling tailored for high-vibration flight environments.
- Integrated a multi-sensor subsystem, interfacing a 6-DOF IMU, high-G accelerometer, barometric altimeter, and multi-constellation GNSS with a 1 Gbit dual-die Quad-SPI Flash memory for non-volatile flight data logging.

Hardware Engineering Intern, Altitut **June 2026 – Aug 2026**

- Architected a scalable, edge-AI hardware platform for real-time video processing, defining the core SoC (Rockchip), power management, and peripheral schematics to locally host the Gemma 4 12B Unified AI Model.
- Designed and routed a 2-layer Raspberry Pi CM5 control board PCB featuring 90-ohm impedance-controlled differential pairs for MIPI/USB-C flashing, plus battery protection and charging circuitry.

FPGA Design Intern, MIT Beaver Works Summer Institute **May 2024 – Aug 2024**

- Collaborated in a 5-person team to architect a custom ASIC and PCB for energy-efficient LED matrix control, utilizing Verilog HDL for RTL design and logic synthesis.
- Optimized control overhead to achieve a 30% power reduction through custom Verilog-based SPI design.
- Executed a complete RTL-to-GDSII flow using OpenLane targeting the SkyWater 130nm node, including 3D layout inspection and Verilog logic synthesis.

Projects

Level 2 Solar-Powered EV Charger

- Engineered a solar-powered Tesla EV charger in a three-person team, integrating eight 240W solar panels wired in series, a 48V battery system, an AC/DC inverter, and a 2500W MPPT charge controller to deliver 2.4 kWh.
- Coordinated local sponsor outreach, resulting in a $4,000 technical supply donation.

Custom FPV Drone

- Designed and fabricated a custom FPV drone from scratch using open-source components, integrating a live-feed video transmitter and radio for remote communication.
- Utilized Fusion360 and Solidworks to engineer 3D-printed landing stilts, propeller guards, and a custom claw, earning 1st place at the TSA Texas State competition.

SkateCam

- Created a multiplayer skating game utilizing Computer Vision to process physical hand inputs via OpenCV and MediaPipe; mapped nodes on users' hands to enable real-time, head-to-head gesture control.
- Received 1st prize at Texas A&M's 2025 HowdyHack Hackathon.

Skills

Python, MATLAB, Verilog HDL, OpenLane, Altium Designer, C++, Docker, SolidWorks, Fusion 360, Autodesk Inventor, Ultimaker Cura, Betaflight